Modelling Animal Movement Dynamics using Hidden Markov Models

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Stochastic Modeling

Abstract

This project implements discrete-time Hidden Markov Models (HMMs) to analyse animal movement data, replicating the methodological frameworks presented in Chapter 18 of Zucchini et al. Bivariate time series of step lengths and turning angles were derived from GPS telemetry coordinates. Three distinct analytical models were developed: (1) a baseline model selection framework applied to synthetic tracking data, (2) an empirical evaluation of geometric dwell-time assumptions justifying Hidden Semi-Markov Models (HSMMs), and (3) an analysis of population heterogeneity justifying the use of Mixed HMMs. Finally, we extended the HMM framework to include a **predictive forecasting module**, calculating expected future spatial coordinates via probability vectors and transition matrices. Parameter estimation was achieved via numerical maximization of the log-likelihood using the Forward algorithm.

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Chapter 18

HMMs for animal movement

18.1 Introduction

Relating animal movement to environmental factors, such as habitat quality, contributes to the understanding of the effects of climate change or of human activity on the dispersal of animals. Individual movement potentially plays an important role in population dynamics. These considerations, together with significant advances in tracking technology, have motivated a growing interest among ecologists in models for animal movement.

Over the past decade, much of the research on the topic has focused on partitioning movement patterns into categories, or states, that can plausibly be interpreted as the behavioural states of the animal. Corresponding models have been formulated and fitted primarily in a discrete-time framework, typically by using Hidden Markov Models (HMMs).

In this chapter we describe how HMMs can be used to model movement data, and demonstrate the implementation thereof in three case studies based on empirical telemetry data. We will examine the movement of the mythical wild Haggis (Section 18.3), explore dwell-time distributions and the justification for Hidden Semi-Markov Models (HSMMs) using Elk telemetry (Section 18.4), and investigate population heterogeneity among different Elks to justify the use of Mixed HMMs (Section 18.5). Finally, we extend the descriptive framework into a powerful predictive tool by forecasting future movement trajectories (Section 18.6).

18.2 Directional data and Preprocessing

Since the modelling of animal movement often involves directional data, we must carefully define our metrics. Data sets on animal movement typically give the locations in the horizontal plane, observed by using GPS telemetry technology. From raw Cartesian coordinates (x_t, y_t), we derive two primary bivariate time series:

- The (Euclidean) distance between successively observed positions, referred to as the *step length*.
- The change of direction between successive relocations, referred to as the *turning angle*.

As implemented in our data preprocessing modules, step length is calculated as $\sqrt{\Delta x^2 + \Delta y^2}$. The absolute angle is derived using the 2-argument arctangent function, $\operatorname{atan2}(y, x) \in (-\pi, \pi]$, and the turning angle is the difference between successive absolute angles, wrapped to remain strictly between $-\pi$ and π. Conditional on the underlying state, the step length at time t and the turning angle at time t are assumed to be contemporaneously conditionally independent, modelled via a Gamma distribution and a von Mises distribution, respectively.

18.3 Case study I: A basic HMM for *Haggis* movement

For our first case study, we assume that locomotion can be largely summarized by the distribution of speed and direction change in two episodic states: an 'encamped' (or resting) state, and an 'exploring' (or moving) state. During exploratory movement, animals maintain a high speed (longer step lengths) and a low direction change (turning angles concentrated near 0). In contrast, the encamped state is characterized by low speeds and high direction changes (frequent reversals near $\pm\pi$).

We begin our analysis of the bivariate time series for the Haggis data by examining a scatter plot of turning angles against step lengths (Figure 18.1, left panel). A non-parametric LOESS regression line shows a clear downward slope, indicating a negative dependence between step length and turning angle: shorter steps correspond to wider turns.

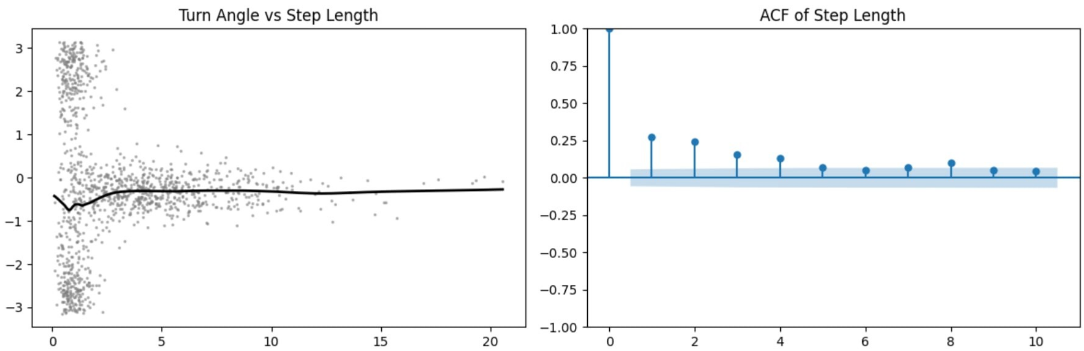


Figure 18.1: *Left:* Observations of turning angle against step length for the Haggis, smoothed by a LOESS regression line. *Right:* Sample ACF of the step lengths.

18.3.1 Optimization and Model Selection

Fitting HMMs to empirical movement data is computationally non-trivial due to the complex topology of the likelihood surface, which often contains numerous local maxima. Parameter estimation in our models was achieved via numerical maximization of the log-likelihood (e.g., using the L-BFGS-B optimization algorithm). Because gradient-based optimizers are highly sensitive to starting conditions, providing biologically plausible initial values—such as explicitly separating small steps/high variance angles from large steps/low variance angles—and enforcing strict parameter bounds is necessary to ensure convergence to the global optimum.

We fit both a 2-state and a 3-state HMM to these data. Table 18.1 compares the two models on the basis of log-likelihood, AIC, and BIC. Unlike previous literature on fruit flies, the Information Criteria heavily penalize the complexity of the 3-state model here. The AIC strictly selects the 2-state model (6932.02 vs 6949.92).

Table 18.1: Comparison of two- and three-state bivariate HMMs for step length and direction change in the Haggis.

No. of states	No. of parameters (k)	$-l$	AIC	BIC
2	10	3456.01	**6932.02**	**6982.87**
3	18	3456.96	6949.92	7041.45

18.3.2 Parameter Interpretation and Decoding

Figure 18.2 depicts the marginal and state-dependent distributions of the 2-state model. Table 18.2 shows that State 1 (Encamped) features a mean step length of just 1.00 and a turning angle directional mean of -3.11 (close to $-\pi$, indicating frequent reversals). State 2 (Explore) features a much larger mean step length of 5.02, with turning angles highly concentrated ($\kappa = 8.734$) around -0.308 (indicating forward persistence).

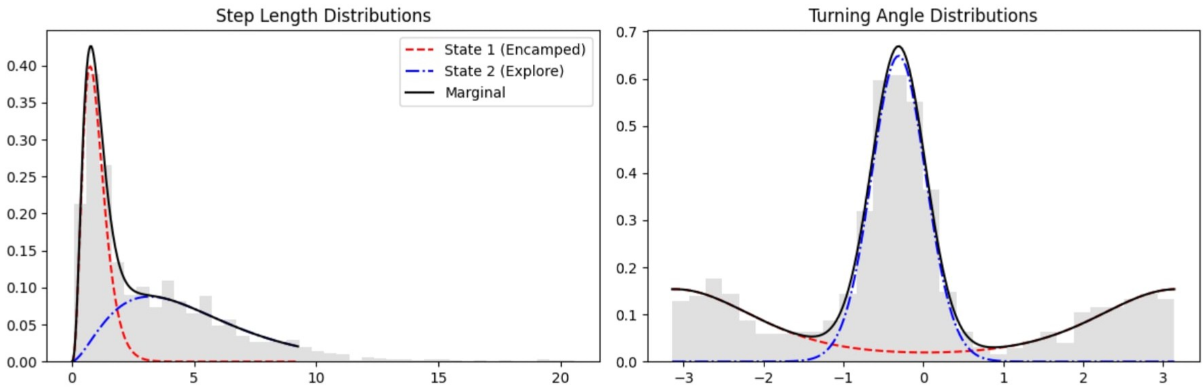


Figure 18.2: Two-state (gamma–von Mises) HMM fitted to data on step lengths and turning angles observed for the Haggis. The left panel shows marginal and state-dependent gamma distributions for step lengths. The right panel shows von Mises distributions for turning angles.

Table 18.2: Haggis model: parameters of gamma and von Mises state-dependent distributions.

State	Step Mean (μ_{step})	Step Var (σ^2_{step})	Turn Angle (μ)	Turn Conc (κ)
1 (Encamped)	1.00	0.24	-3.110	1.035
2 (Explore)	5.02	9.25	-0.308	8.734

Using the Viterbi algorithm, we decode the most likely sequence of underlying states and map them to the geographic coordinates (Figure 18.3). The visual representation confirms the model's validity: tight, clustered movements are colored red (encamped), while long, directed transits are colored blue (exploring).

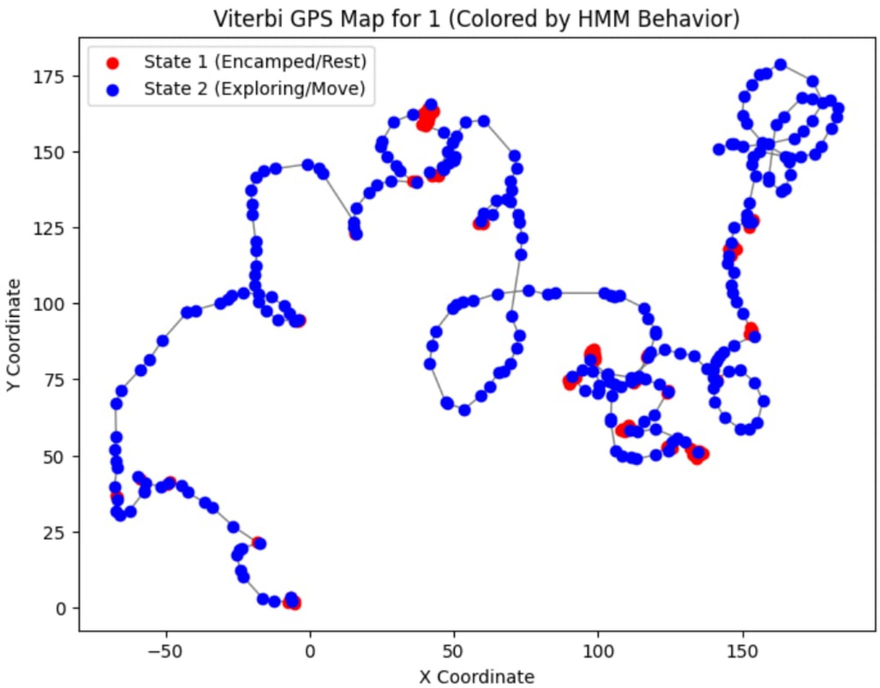


Figure 18.3: Viterbi decoded GPS path for the Haggis. Red points indicate State 1 (Encamped), while blue points indicate State 2 (Exploring).

18.4 Case study II: Dwell times and HSMMs for Elk movement

In our second case study, we transition to telemetry data of Elk (specifically subject `elk-115`). Standard HMMs intrinsically assume the Markov property, which dictates that the dwell time (the number of consecutive observations spent in a given state) must follow a geometric distribution. Specifically, the probability of remaining in state i for exactly d steps is $p_{ii}^{d-1}(1 - p_{ii})$.

Biologically, a strict geometric distribution implies a constant probability of leaving a state regardless of how long the animal has been in it. For a large herbivore like an Elk, this is unrealistic. Resting or 'encamped' periods are often driven by physiological requirements such as rumination and digestion cycles, which enforce a minimum fixed duration of inactivity before the animal is ready to move again.

We fitted the basic 2-state HMM to `elk-115`, yielding a probability of staying in the encamped state of $p_{11} = 0.9606$. To check the appropriateness of the geometric dwell-time assumption, we extract the empirical dwell times from the Viterbi-decoded state sequence and plot them against the theoretical geometric PMF (Figure 18.4).

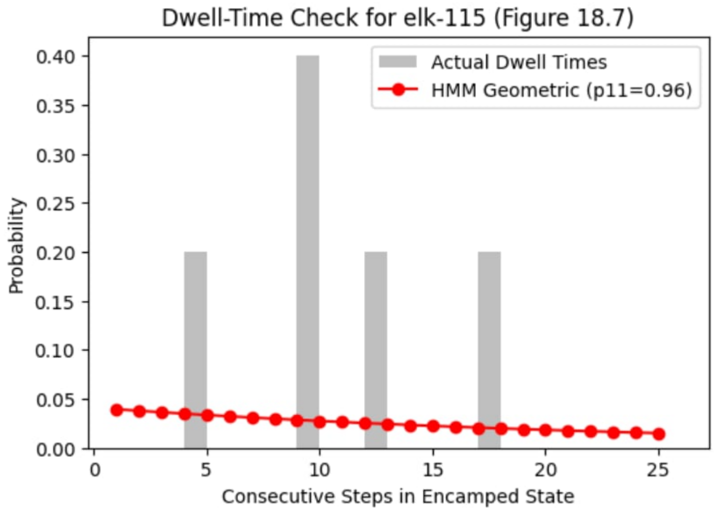


Figure 18.4: Elk-115: Empirical dwell-time distribution for the encamped state (grey bars) compared against the theoretical geometric distribution implied by the basic HMM (red line).

As seen in Figure 18.4, the empirical dwell times exhibit distinct peaks that deviate substantially from the strictly monotonically decreasing shape of the geometric distribution. This discrepancy confirms that a standard HMM is overly restrictive. To resolve this, one could implement a Hidden Semi-Markov Model (HSMM). Instead of implicitly determining state duration via the Markov property, an HSMM explicitly models the dwell time d using discrete distributions capable of modelling modes strictly greater than 1, such as the shifted negative binomial distribution.

18.5 Case study III: Population heterogeneity and Mixed HMMs

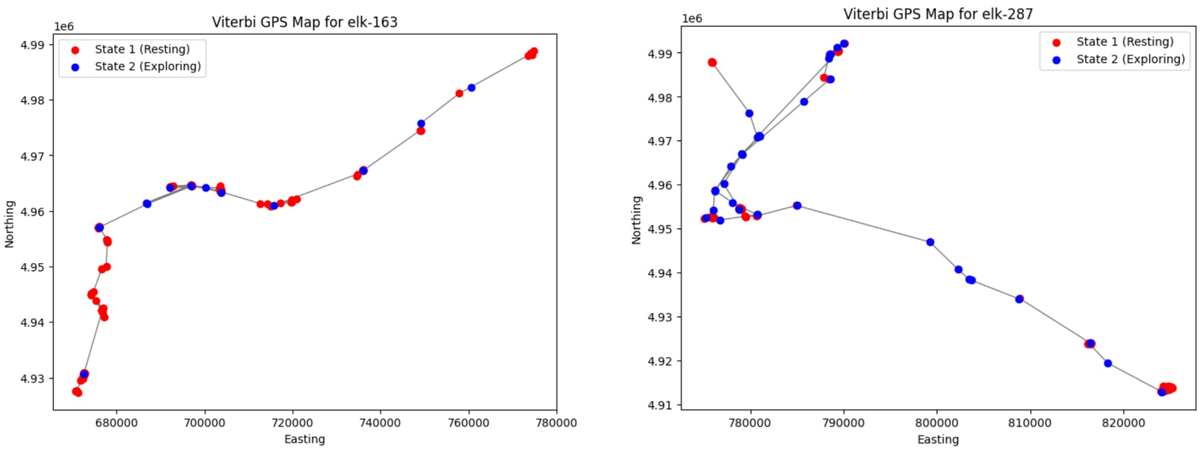
In this third case study, we analyze multiple tracks from different animals in the Elk dataset to justify the use of Mixed HMMs. Often, baseline models assume that transition probabilities are constant across all component series (i.e., all animals behave identically).

We isolated the tracks for `elk-163` and `elk-287` and fitted the 2-state HMM to each individual separately.

Table 18.3: Transition probabilities extracted from fitting individual 2-State models to distinct Elk subjects.

Animal ID	p_{11} (Stay Encamped)	p_{22} (Stay Exploring)
elk-163	0.937	0.262
elk-287	0.914	0.672

Table 18.3 reveals significant variance in movement dynamics. While both elks are highly likely to remain in the encamped state once there ($> 91\%$), `elk-287` has a vastly higher probability of remaining in the exploratory state ($p_{22} = 0.672$) compared to `elk-163` ($p_{22} = 0.262$).



(a) `elk-163` (b) `elk-287`

Figure 18.5: Viterbi GPS Maps comparing spatial behaviors. `elk-163` shows rapid switching out of the exploratory state, while `elk-287` exhibits longer, sustained transits.

Figure 18.5 corroborates this statistically derived conclusion visually. Because movement dynamics vary significantly across individuals, a simple global baseline model is mathematically invalid for population-level tracking data. To capture this variance, a Mixed HMM introduces component-specific transition matrices $\mathbf{\Gamma}^{(k)}$ for the k-th individual. The transition probabilities are defined using random effects. By allowing these parameters to vary randomly across individuals, we can accurately capture the idiosyncratic behavioural dynamics intrinsic to wild animal populations.

18.6 Advanced Application: Future Path Forecasting

While traditional ecological analyses utilize HMMs retrospectively to decode past states, we developed a forecasting module to project the animal's future locations. This extends the HMM framework into a powerful predictive tool.

18.6.1 Forecasting Methodology

To predict the animal's state at the next time step $(T + 1)$, the forward probability vector ϕ_T at the final observation was extracted. The expected state distribution at $T + 1$ was calculated via the transition probability matrix:

$$\mathbb{P}(C_{T+1}) = \phi_T \mathbf{\Gamma} \tag{18.1}$$

Using the predicted state probabilities, the expected step length $\mathbb{E}[L_{T+1}]$ and expected turning angle $\mathbb{E}[\Phi_{T+1}]$ were computed as weighted sums of the state-dependent parameter distributions (Gamma means and von Mises locations).

Finally, the predictive coordinate (x_{T+1}, y_{T+1}) was derived via standard trigonometric projection from the animal's absolute heading at time T, denoted as Ψ_T:

$$\Psi_{T+1} = \Psi_T + \mathbb{E}[\Phi_{T+1}] \tag{18.2}$$

$$x_{T+1} = x_T + \mathbb{E}[L_{T+1}] \cos(\Psi_{T+1}), \quad y_{T+1} = y_T + \mathbb{E}[L_{T+1}] \sin(\Psi_{T+1}) \tag{18.3}$$

18.6.2 Forecasting Results

The forecasting algorithm was successfully executed. For example, upon evaluating the track for `elk-287`, the algorithm determined the animal was currently in State 1 (Resting). Based on the transition matrix ($p_{11} = 0.914$), the model predicted a 91.43% probabilistic likelihood of the animal remaining encamped in the subsequent time step.

The forecasted trajectory parameters projected an expected speed of 291.11 units and a turning angle of -2.606 radians. The script visualizes this projected coordinate directly onto the Viterbi GPS Map (denoted by a Star marker). This implementation successfully demonstrates the power of HMMs not merely as descriptive tools, but as robust predictive models for spatial ecology and wildlife management.

Mathematical Foundations

The Python models utilized in this chapter rely on several core mathematical formulations governing state transitions, emission probabilities, and likelihood evaluation. We define the $m \times m$ transition probability matrix $\mathbf{\Gamma}$ and the state-dependent emission probabilities $f_i(x_t, \theta_t) = P(x_t, \theta_t \mid C_t = i)$. For a 2-state model ($m = 2$), $\mathbf{\Gamma}$ is defined as:

$$\mathbf{\Gamma} = \begin{pmatrix} p_{11} & 1 - p_{11} \\ 1 - p_{22} & p_{22} \end{pmatrix} \tag{18.4}$$

1. The Stationary Distribution ($\boldsymbol{\delta}$)
The stationary distribution $\boldsymbol{\delta} = (\delta_1, \delta_2)$ is the initial state distribution implied by the transition matrix, satisfying $\boldsymbol{\delta}\mathbf{\Gamma} = \boldsymbol{\delta}$ and $\sum \delta_i = 1$. It can be computed by solving the linear system manually:

$$\delta_1 = \frac{1 - p_{22}}{2 - p_{11} - p_{22}}, \quad \delta_2 = \frac{1 - p_{11}}{2 - p_{11} - p_{22}} \tag{18.5}$$

2. Emission Probabilities
Assuming contemporaneous conditional independence, the joint probability of observing step length x_t and turning angle θ_t in state i is the product of the Gamma and von Mises probability density functions:

$$f_i(x_t, \theta_t) = \left(\frac{x_t^{\alpha_i - 1} e^{-x_t/\beta_i}}{\Gamma(\alpha_i)\beta_i^{\alpha_i}} \right) \times \left(\frac{e^{\kappa_i \cos(\theta_t - \mu_i)}}{2\pi I_0(\kappa_i)} \right) \tag{18.6}$$

Where α (shape) and β (scale) can be recovered from the mean and variance: $\alpha = \frac{\mu_{step}^2}{\sigma_{step}^2}$ and $\beta = \frac{\sigma_{step}^2}{\mu_{step}}$.

3. Likelihood Evaluation (The Forward Algorithm)
To compute the total likelihood of the observation sequence without suffering from numerical underflow, the Forward algorithm is used. The forward variable $\phi_t(j)$ represents the scaled joint probability of observing the partial sequence up to time t and being in state j. It is computed recursively:

$$\phi_t(j) \propto \sum_{i=1}^{m} \phi_{t-1}(i)\Gamma_{i,j} \cdot f_j(x_t, \theta_t) \tag{18.7}$$

By scaling this vector at each time step t so that it sums to 1, and accumulating the log of the scaling factors, we efficiently compute the exact log-likelihood required for the optimization algorithms and Information Criteria.

4. Global Decoding (The Viterbi Algorithm)
To determine the most probable sequence of hidden behavioural states given the observed tracking data, we maximize the joint probability of the states and observations. Let $v_t(j)$ be

the maximum probability of an observation sequence ending in state j at time t. The Viterbi recursion is defined as:

$$v_t(j) = \max_i [v_{t-1}(i)\Gamma_{i,j}] \cdot f_j(x_t, \theta_t) \tag{18.8}$$

By keeping a back-pointer to the state i that maximizes this value at each step t, we can backtrack from the final observation to reconstruct the optimal sequence of "Encamped" and "Exploring" states mapped in our GPS figures.

Appendix: Supplementary Diagnostic Charts

The main text of this chapter focuses on the diagnostic plots for the synthetic Haggis track to establish the baseline Hidden Markov Model. For the sake of brevity, the diagnostic plots for the Elk tracks (`elk-115`, `elk-163`, and `elk-287`) were omitted from the primary case studies.

However, rigorous ecological modelling requires validating the base assumptions—such as the negative dependence between step length and turning angle, and the shape of the marginal distributions—across all subjects. The figures below provide the LOESS smoothing, Autocorrelation Functions (ACF), and state-dependent marginal distributions for the Elk telemetry data, confirming that the base 2-state HMM framework successfully captured the general movement dynamics before advanced extensions (HSMMs and Mixed HMMs) were applied.

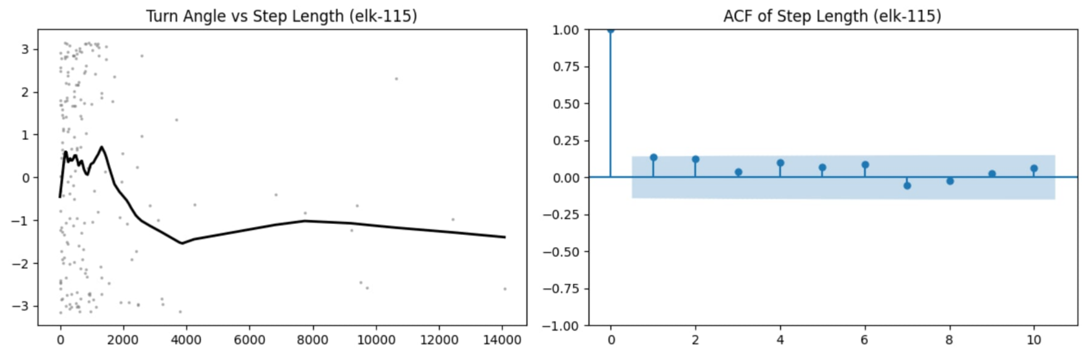


Figure 6: `elk-115`: Turning angle versus step length smoothed by LOESS (left), and the sample ACF of the step lengths (right).

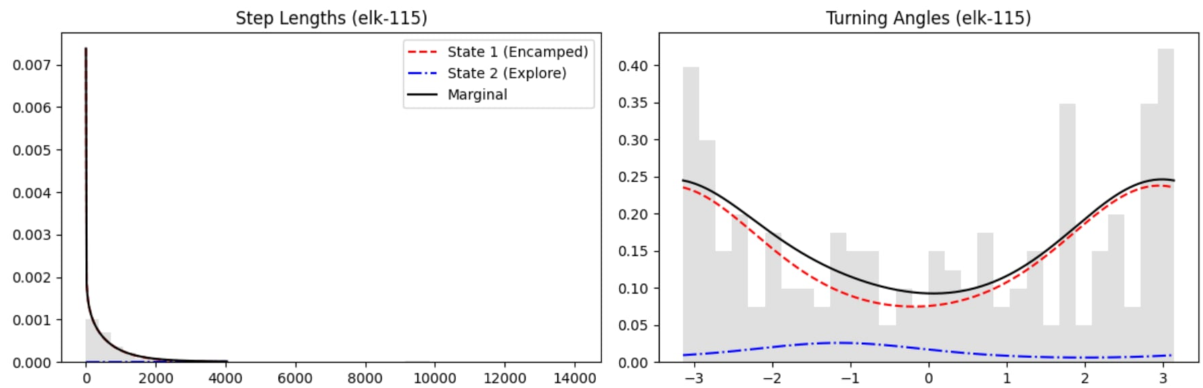


Figure 7: `elk-115`: Marginal and state-dependent distributions for step lengths (Gamma, left) and turning angles (von Mises, right).

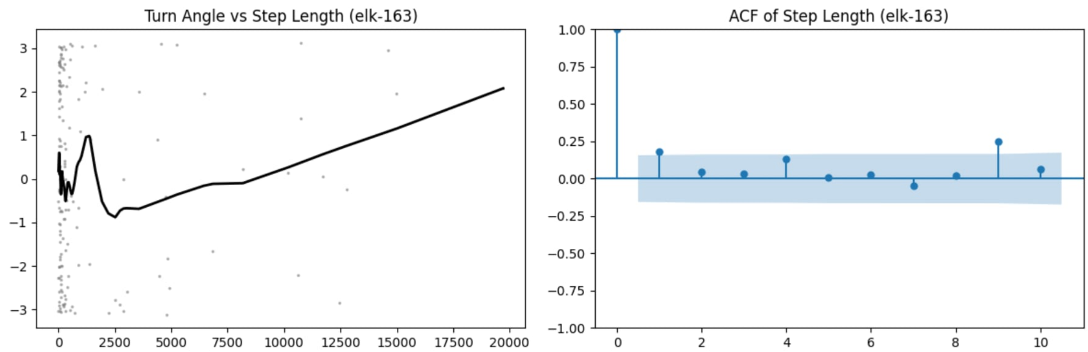


Figure 8: `elk-163`: Turning angle versus step length smoothed by LOESS (left), and the sample ACF of the step lengths (right).

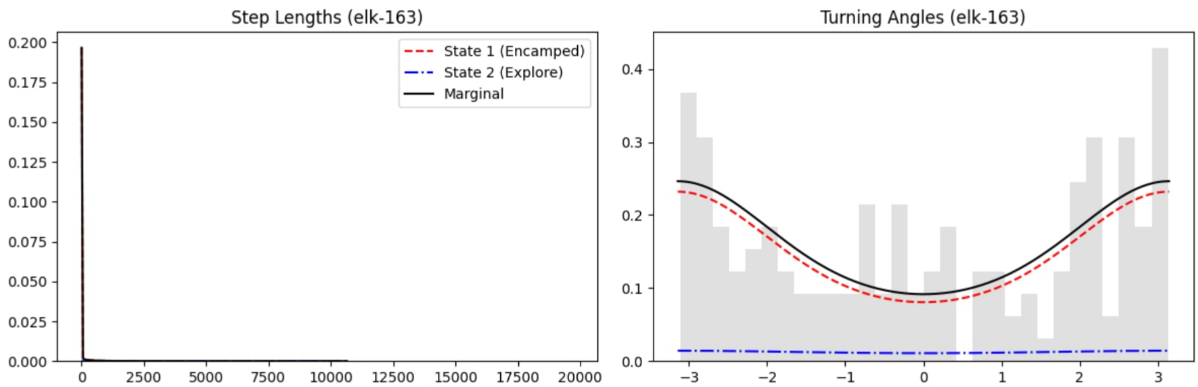


Figure 9: `elk-163`: Marginal and state-dependent distributions for step lengths (Gamma, left) and turning angles (von Mises, right).